

Topic 6b - Algebra of Taxes (& subsidies)



Demand: $Q_D = Q_D(p^d)$

p^d Price paid by consumers

Supply: $Q_S = Q_S(p^s)$

p^s Price received by sellers

In the absence of any restriction $\Rightarrow$ Mkt eq: $p^e = p^d = p^s$

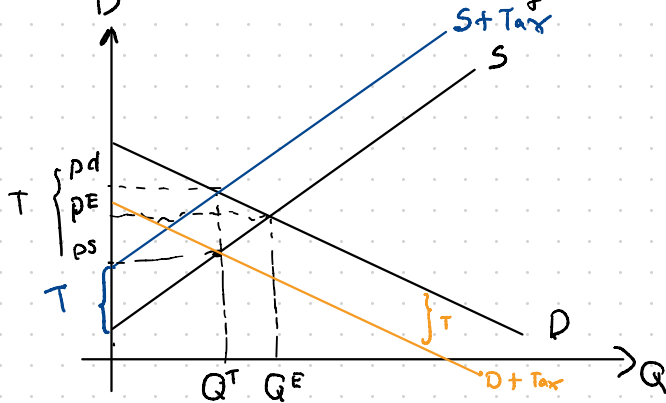
$$Q_D(p^e) = Q_S(p^e) \Rightarrow Q^E$$

When there is a per-unit tax: introduces a wedge between p^d and $p^s \Rightarrow p^d \neq p^s$

The eq. relating them is $T = p^d - p^s$, this together with

$$Q_D(p^d) = Q_S(p^s)$$

determines the new eq. after tax: (Q^T, p^d, p^s)



Assume a Linear Demand & Supply

$$Q_D = a - bP^D, \quad Q_S = c + dP^S$$

Per-unit tax is $T \Rightarrow T = P^D - P^S$

① $P^S = P^D - T \rightarrow$ "Auxiliary shift of the S curve"

② $P^D = T + P^S \Rightarrow$ "Auxiliary shift of the D curve"

Follow ① $\Downarrow$

$$Q_S = c + d(P^D - T)$$

$$= c - dT + dP^D$$

To obtain P^D, P^S, Q^T we solve:

$$a - bP^D = c - dT + dP^D$$

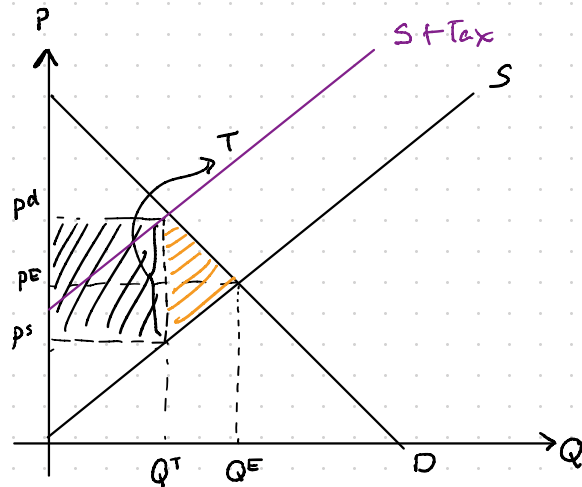
$$\Rightarrow \boxed{P^D = \frac{a - c + dT}{b + d}}$$

$$P^S = P^D - T = \frac{a - c + dT}{b + d} - T = \frac{a - c - bT}{b + d} = P^S$$

$$Q^T = a - b \left(\frac{a - c + dT}{b + d} \right) = \frac{ad + bc - bdT}{b + d}$$

$$P^E = \frac{a - c}{b + d}; \quad Q^E = \frac{ad + bc}{b + d}, \quad \text{when } T = 0$$

(P^E, Q^E) free-market equilibrium price & quantity



$$\text{Government Revenue} = G = T \cdot Q^T = (P^D - P^S)Q^T = \boxed{\frac{ad + bc - bdT}{b + d} \cdot T}$$

$$DWL = \boxed{\frac{(Q^E - Q^T) \times T}{2}} = \frac{bdT \times T}{2(b+d)} = \frac{bdT^2}{2(b+d)}$$

$$DWL \propto bdT^2$$

$\uparrow$
"all proportional"

Tax Incidence: $p^E = \frac{a-c}{b+d}$

Consumers	Incidence
$p_d = \frac{a-c+dT}{b+d}$	$\frac{p_d - p^E}{T} = \frac{\frac{a-c+dT}{b+d} - \frac{a-c}{b+d}}{T}$ $= \frac{\frac{dT}{b+d}}{T} = \frac{d}{b+d} = \underline{TI_D}$
Producers	
$p_s = \frac{a-c-bT}{b+d}$	$\frac{p^E - p_s}{T} = \frac{\frac{a-c}{b+d} - \frac{a-c-bT}{b+d}}{T}$ $= \frac{\frac{bT}{b+d}}{T} = \frac{b}{b+d} = \underline{TI_S}$

$$TI_D > TI_S \Leftrightarrow \frac{TI_D}{TI_S} > 1 \Leftrightarrow \frac{\frac{d}{b+d}}{\frac{b}{b+d}} > 1$$

$$\Leftrightarrow \frac{d}{b} > 1 \Leftrightarrow d > b$$

$$TI_D > TI_S \Leftrightarrow d > b$$

Recall the point elasticity formulas

$$\epsilon_p^S = d \cdot \frac{P}{Q_s}, \quad |\epsilon_p^D| = b \cdot \frac{P}{Q_D}$$

Calculating incidence at the same point (P, Q)

$$d > b \Rightarrow \frac{dP}{Q} > \frac{bP}{Q}$$

Multiply by $\frac{P}{Q}$

$$\Leftrightarrow \epsilon_p^S > |\epsilon_p^D|$$

$$Q_D = 100 - 2P^d, Q_S = 10 + P^s, T = 10$$

Fundamental Relationship: $10 = P^d - P^s \Rightarrow P^s = P^d - 10$

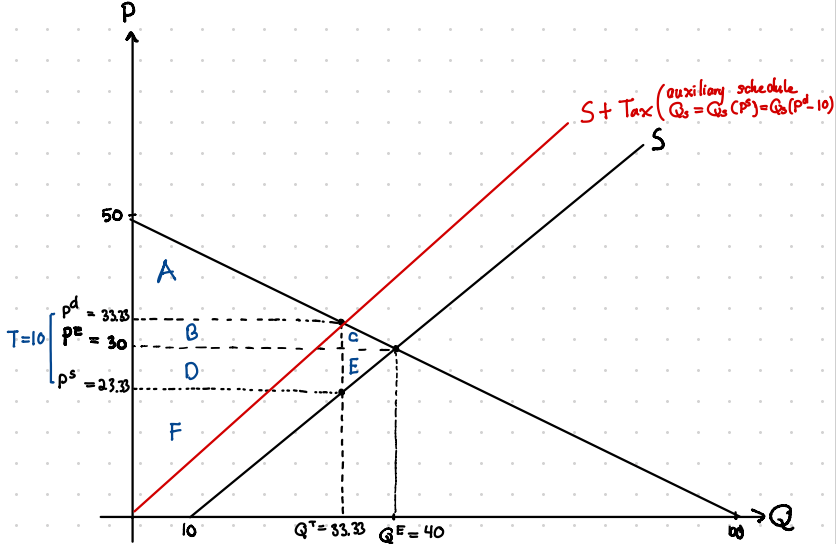
Plug P^s into Q_S : $Q_S = 10 + P^d - 10 = P^d$

Equate Q_D with Q_S : $100 - 2P^d = P^d \Rightarrow P^d = \frac{100}{3} \approx 33.33$

$P^s = \frac{100}{3} - 10 \approx 23.33$; $Q^T = 10 + 23.33 \approx 33.33$

If $T=0$; $P^d = P^s = P^E = \frac{100-10}{2+1} = 30$

$Q^E = 40$



	Area	Amount
Tax Paid by consumers	B	$(P^d - P^E)Q^T = (33.33 - 30)(33.33) \approx 110.44$
Tax Paid by producers	D	$(P^E - P^s)Q^T = (30 - 23.33)(33.33) \approx 222.34$
Total Gov. Tax Collection	B + D	$(P^d - P^s)Q^T = T \cdot Q^T = 10 \cdot 33.33 \approx 333.33$

	Consumer Surplus	Producer Surplus
Before Tax	Area = $ABC = \frac{40(50-30)}{2} = 400$	Area = $DEF = \frac{1}{2}(40-10)(30) + 10 \times 30 = 750$
After Tax	Area = $A = \frac{1}{2}(50-23.33)(33.33) \approx 277.81$	Area = $F = \frac{1}{2}(23.33-10)(23.33) + 10 \times 23.33 = 505.47$

$$DWL = C + E = \frac{1}{2} T (40 - 33.33) = \frac{10}{2} (40 - 33.33) = 33.35$$

$$SS^{\text{before}} = 1150$$

$$SS^{\text{after}} = 783.28$$

$$R = 333.33$$

$$DWL = 33.35$$

$$SS^{\text{before}} = SS^{\text{after}} + R + DWL$$

$$\Delta SS = \text{change in Social Surplus} = R + DWL$$

- ⚠ Change in Social Surplus is the sum of
- Government Revenues (which can be used on public goods & services (best case → corruption))
 - A deadweight loss that no economic agent appropriates.